Proof of Concept (PoC):

Security Monitoring with Splunk SIEM

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Environment: Home Lab / Docker Simulation

Section 1 - Executive Summary

This Proof of Concept (PoC) demonstrates the ability to ingest, normalize, and analyze security-relevant log data using Splunk SIEM.
The objective of the PoC is to simulate a small-scale SOC monitoring environment, detect basic attack patterns (e.g., SSH brute-force), and showcase alerting and dashboard capabilities.

The environment includes:

• Linux authentication logs (auth.log)
• Web server logs (nginx access + error)
• Custom simulated attack events
• Splunk running in Docker with mounted log sources

The PoC reflects realistic SOC workflows used in security monitoring, analysis, and detection engineering.

Section 2 - Architecture

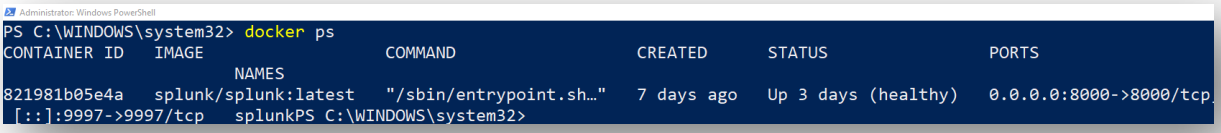
The lab environment consists of a single Splunk server running in a Docker container. The container has a mounted directory (/mnt/logs) that contains simulated Linux and web server log files. Splunk ingests these files through local file monitoring and indexes them for analysis.

Components:
- Splunk Enterprise (Docker)
- Mounted directory: /mnt/logs
- Linux authentication logs (auth.log)
- NGINX error and access logs (error.log, access.log)

Showing the running Splunk container!

Screenshot:

⇨



Section 3 – Data Sources

Log Sources Ingested into Splunk

1. SSH Authentication Logs

• **Purpose:** Failed logins, authentication attempts
• **File Location:** /mnt/logs/auth.log
• **Host:** kali
• **Sourcetype:** linux_auth

2. NGINX Error Logs

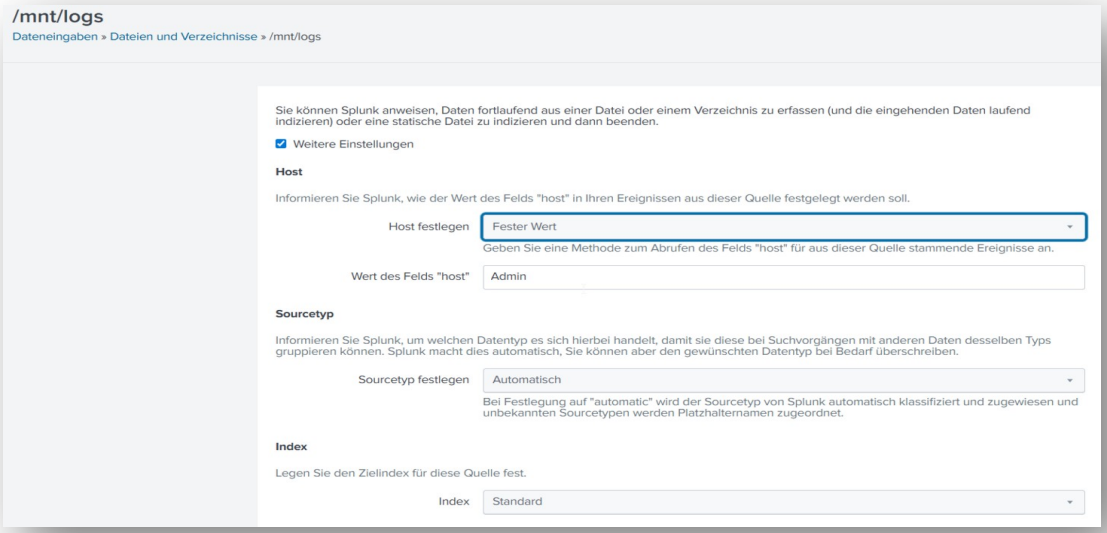
• **Purpose:** Server warnings and error messages
• **File Location:** /mnt/logs/error.log
• **Host:** webserver
• **Sourcetype:** nginx_error

3. NGINX Access Logs

• **Purpose:** HTTP requests and traffic monitoring
• **File Location:** /mnt/logs/access.log
• **Host:** webserver
• **Sourcetype:** nginx_access

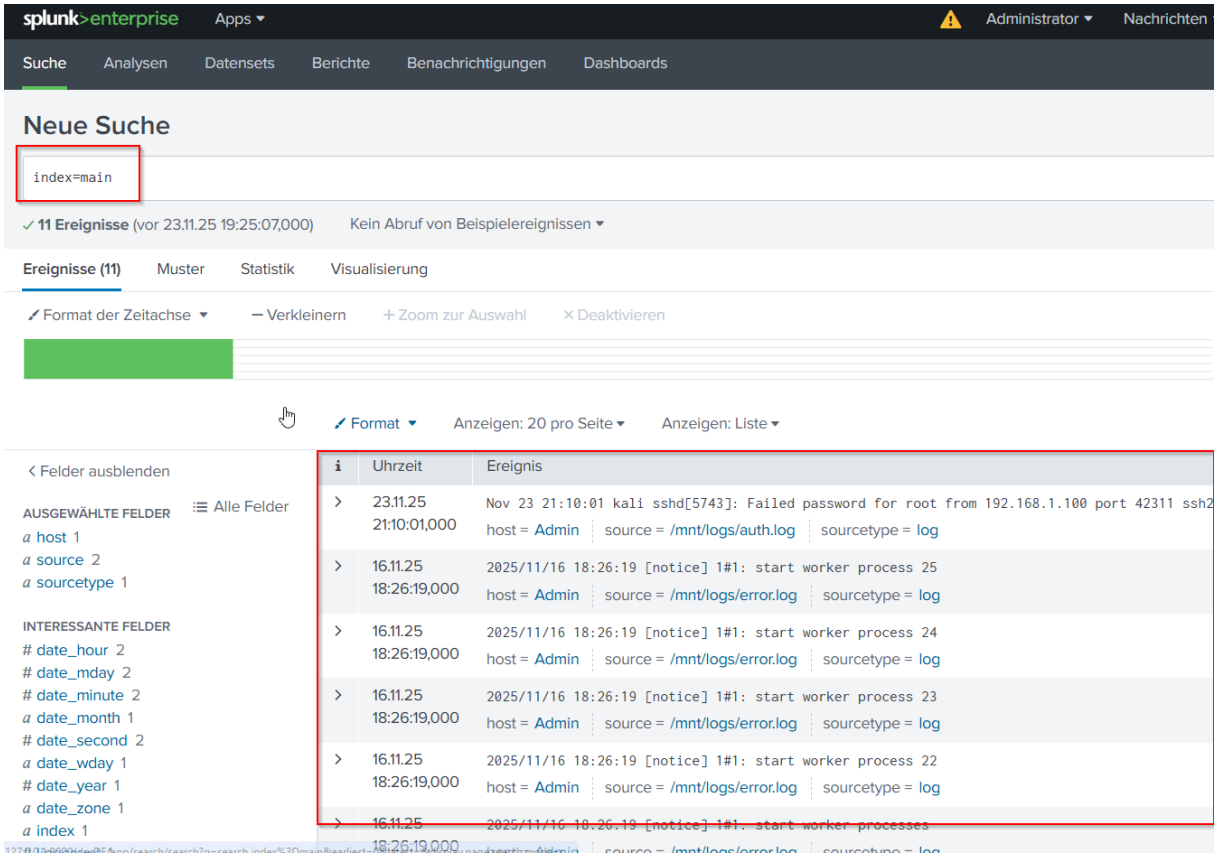
Screenshot:

➔ Data Input settings for /mnt/logs in Splunk



Screenshot:

➔ Indexed events from auth.log, error.log, and access.log



Section 4 — Log Ingestion & Indexing

Splunk was configured to ingest the log files directly from /mnt/logs using the “Files & Directories” input. Each file was assigned an appropriate sourcetype and indexed into the default “main” index. After ingestion, events were verified within Splunk Search to confirm correct timestamp extraction and host assignment.

i	Uhrzeit	Ereignis
>	23.11.25 21:10:01,000	Nov 23 21:10:01 kali sshd[5743]: Failed password for root from 192.168.1.100 port 42311 ssh2 host = Admin | source = /mnt/logs/auth.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: start worker process 25 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: start worker process 24 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: start worker process 23 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: start worker process 22 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: start worker processes host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19) host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: nginx/1.29.3 host = Admin | source = /mnt/logs/error.log | sourcetype = log
>	16.11.25 18:26:19,000	2025/11/16 18:26:19 [notice] 1#1: using the "epoll" event method host = Admin | source = /mnt/logs/error.log | sourcetype = log

Section 5 — Detection Use Case 1: SSH Brute Force Attempts

Objective:

Identify brute-force SSH activity by detecting repeated failed authentication attempts from the same IP address.

SPL Query:

```
index=main "Failed password"
```

This allows us to quickly identify attackers attempting multiple login failures.

Screenshot:

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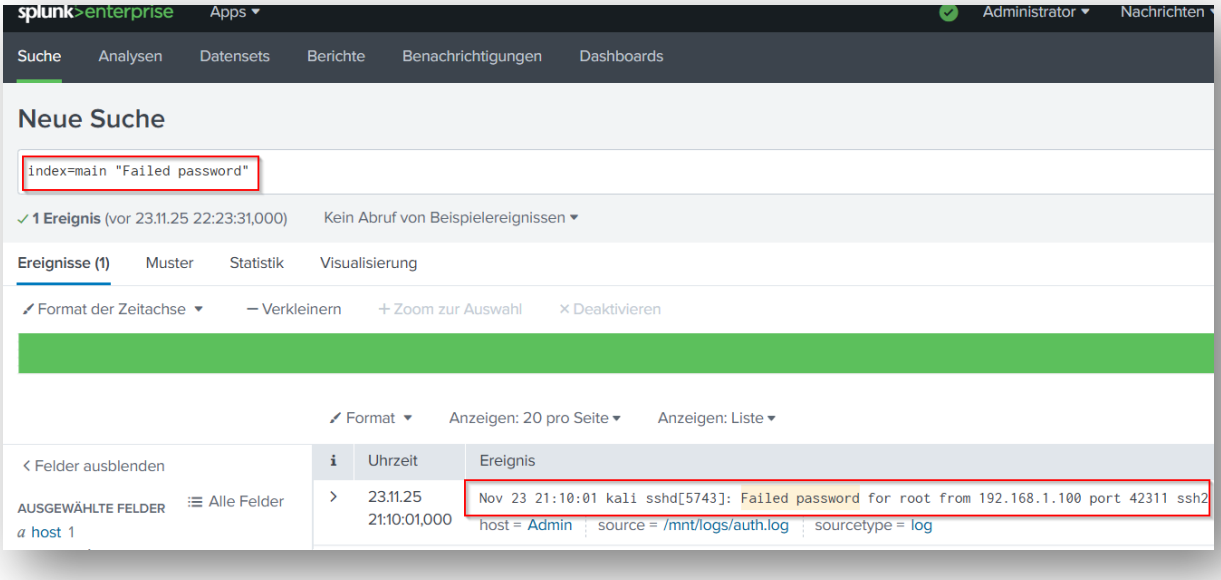


Section 6 — Dashboard Overview

A Splunk dashboard was created to provide a visual overview of SSH failures and web server activity. The dashboard includes:

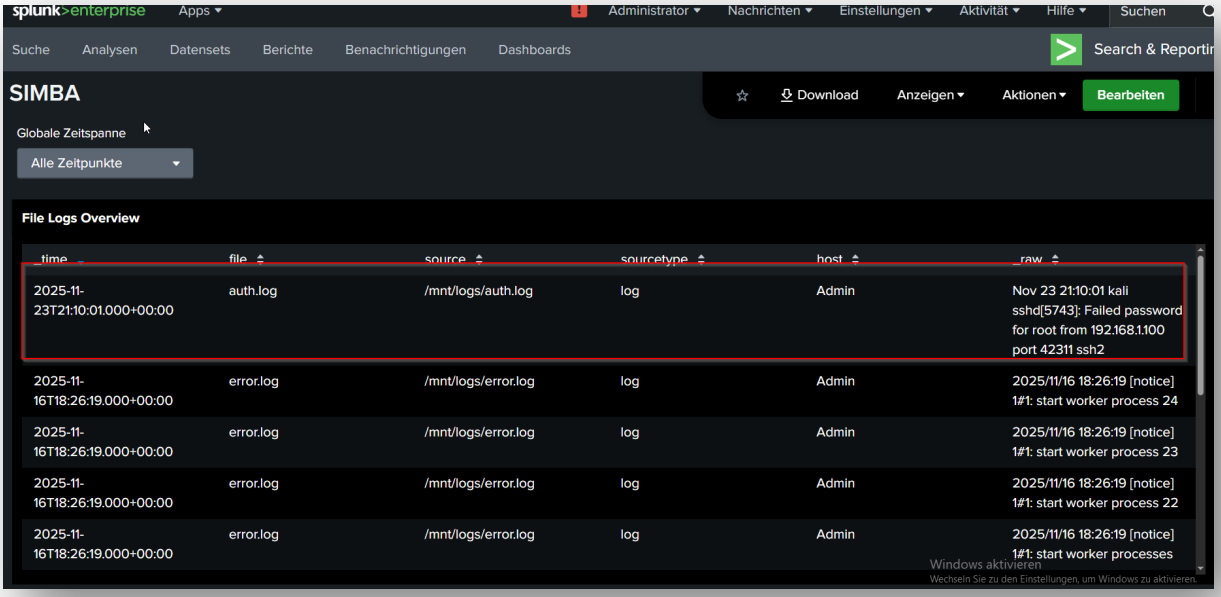
- A chart of failed SSH logins over time

- A panel showing top attacking IPs

- Visualizations of NGINX errors and access patterns

Screenshot

➔



Section 7 — Conclusion

This PoC successfully demonstrates the ingestion, parsing, detection, and visualization capabilities of Splunk SIEM in a simulated environment. The included screenshots validate each step of the configuration and analysis process. This setup can be expanded with additional data sources such as Windows logs, firewall logs, IDS alerts, and automated alerting to create a more advanced SOC monitoring lab.